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Lab - 3

PART - 1

```
#PART - 1

data("airquality")
my_df <- airquality

airquality

str(my_df)
head(my_df,6)

names(my_df)

my_df_temp <- data.frame( temp = c(my_df$Temp))
print(my_df_temp)

length( unique(my_df$Month) )
```

---

1. How many months are in the data set?
   ➔ 5

```
>
> mean_temp <- tapply(my_df$Temp, my_df$Month, mean)
> mean_temp
       5        6        7        8        9
65.54839 79.10000 83.90323 83.96774 76.90000
>
> median_temp <- tapply(my_df$Temp, my_df$Month, median)
> median_temp
 5  6  7  8  9
66 78 84 82 76
>
> sd_temp <- tapply(my_df$Temp, my_df$Month, sd)
> sd_temp
       5        6        7        8        9
6.854870 6.598589 4.315513 6.585256 8.355671
>
```

```
> overall_mean <- mean(my_df$Temp)
> overall_mean
[1] 77.88235
>
> overall_median <- median(my_df$Temp)
> overall_median
[1] 79
>
> overall_sd <- sd(my_df$Temp)
> overall_sd
[1] 9.46527
>
```

PART – 2

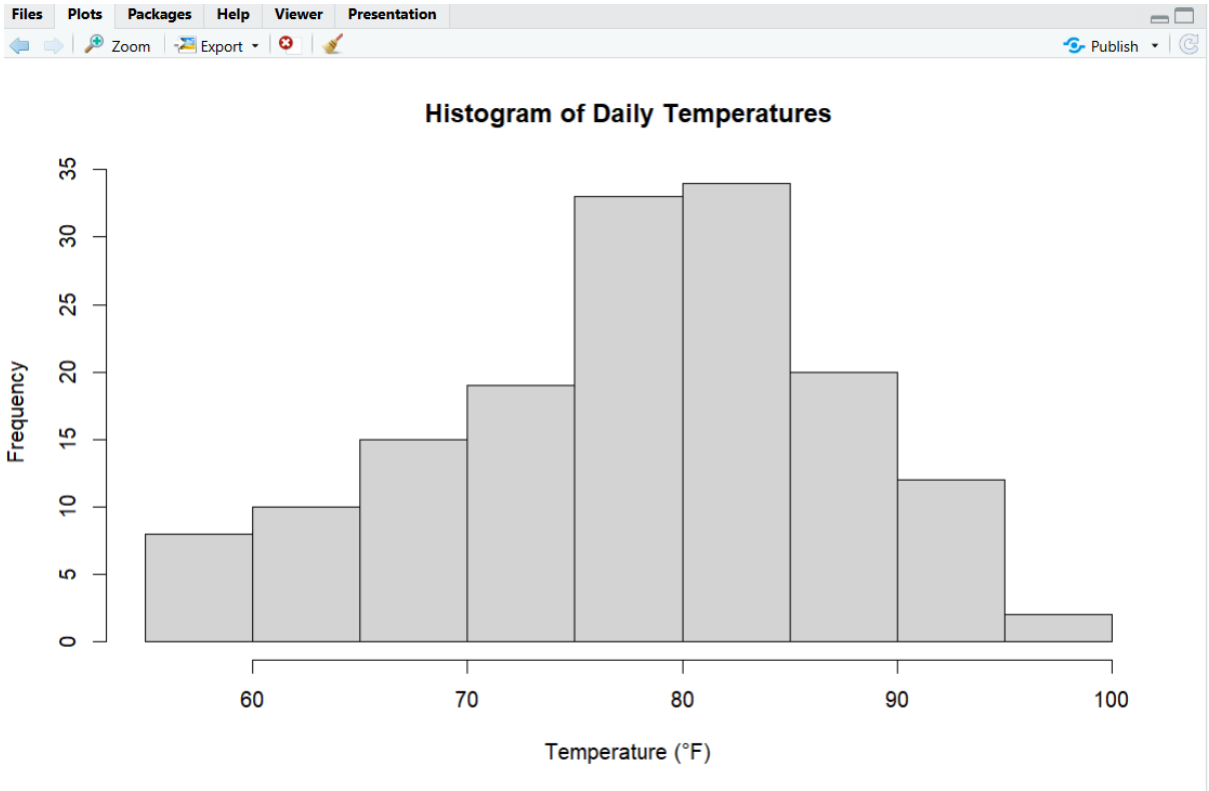
```
#PART - 2

hist(my_df$Temp,
     main = "Histogram of Daily Temperatures",
     xlab = "Temperature (°F)",
     ylab = "Frequency")

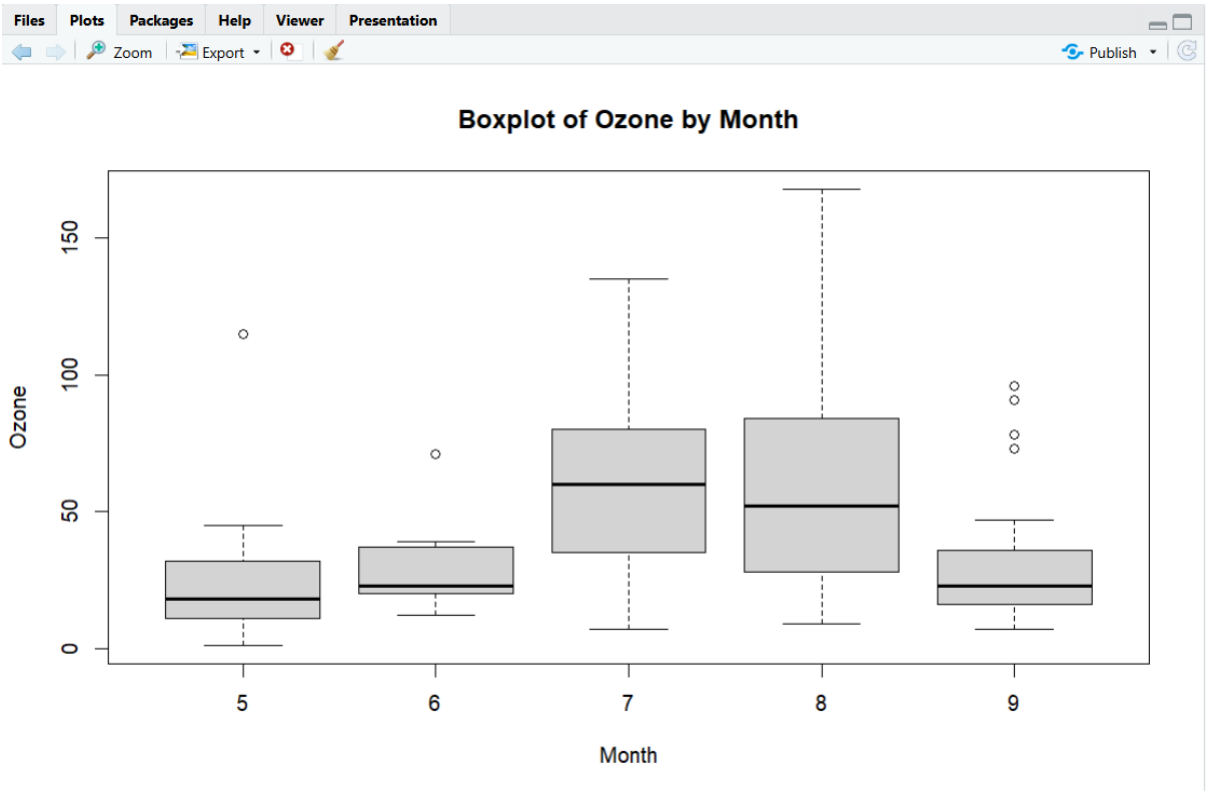
boxplot(Ozone ~ Month, data = my_df,
        main = "Boxplot of Ozone by Month",
        xlab = "Month",
        ylab = "Ozone")

plot(my_df$Temp, my_df$Ozone,
     main = "Scatterplot of Temperature vs. Ozone",
     xlab = "Temperature (°F)",
     ylab = "Ozone" )

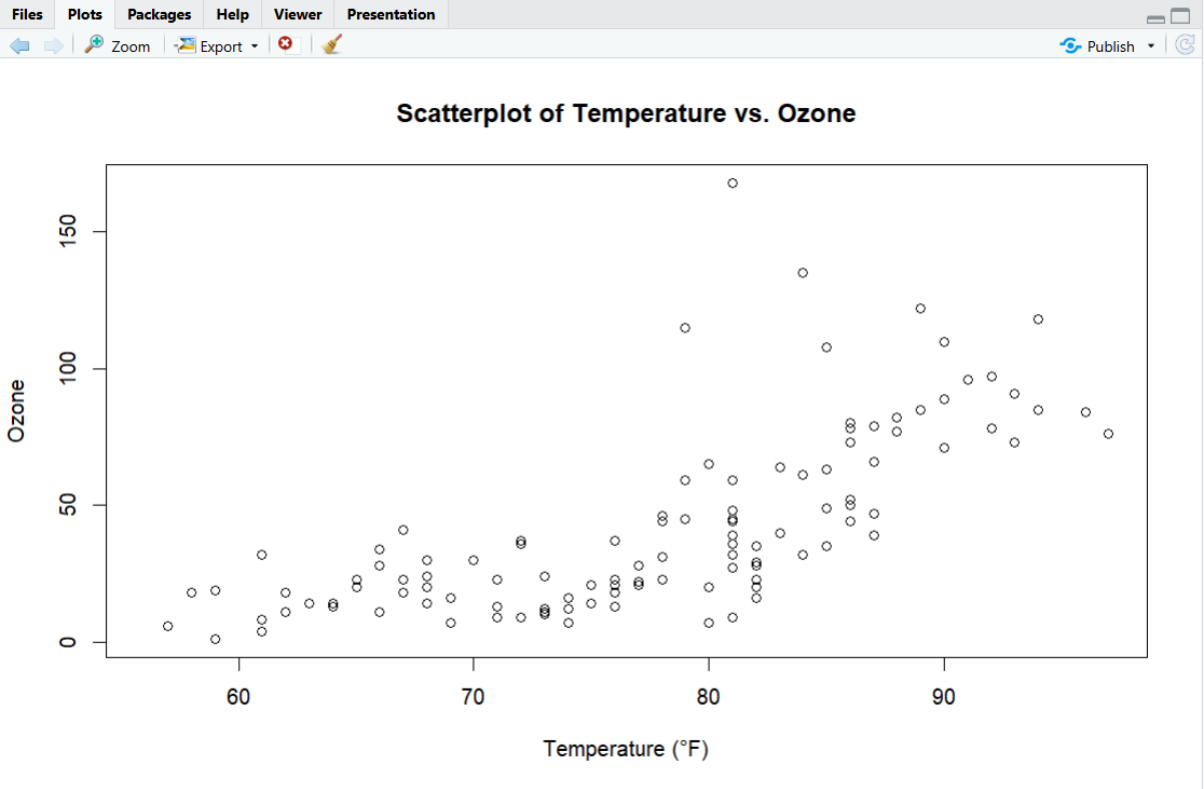
abline(lm(Ozone ~ Temp, data = my_df), col = "blue", lwd = 2)
```



➔ The histogram shows how the daily temperatures are spread. It tells us which temperatures happened most often and which temperatures were less common.



➔ The boxplot shows how Ozone levels are spread for each month. The middle line in each box is the median, the box shows the middle 50% of the data, and the whiskers show the lowest and highest values. Any dots outside the whiskers are outliers. This helps us see which months had higher or lower Ozone levels.



➔ The scatterplot shows the relationship between Temperature and Ozone levels. Each point represents one day. The x-axis is the temperature and the y-axis is the Ozone level. By looking at the plot, we can see if higher temperatures are related to higher or lower Ozone levels.

PART – 3

1. The temperature data is almost symmetric, but the mean is slightly less than the median, so it shows a small left skew.
2. Ozone tends to increase as Temperature increases. The blue regression line shows this trend clearly, making it easier to see the pattern in the scatterplot.

PART – 4

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1. The probability that the temperature during these months is less than 70.

➔ 0.2024884

2. The probability that the temperature during these months is greater than 85.

➔ 0.226033

3. The probability that the temperature during these months is less than 90 and greater than 75.

➔ 0.5194002

➔ Locate line 102 and explain the following. Provide your answers in the submitted PDF. (3 points)

1. which month this is for?

➔ August

2. what the temperature is (in degrees F)?

➔ 92

3. what does the z-score information tell us about this particular temperature value?

➔ 1.491521

```r
#PART - 4

prob_less_70 <- pnorm(70, mean = overall_mean, sd = overall_sd)
prob_less_70

prob_greater_85 <- 1 - pnorm(85, mean = overall_mean, sd = overall_sd)

prob_75_90 <- pnorm(90, mean = overall_mean, sd = overall_sd) -
              pnorm(75, mean = overall_mean, sd = overall_sd)
prob_75_90

mean.Temp.All <- mean(my_df$Temp)
sd.Temp.All   <- sd(my_df$Temp)

my_df$z_score <- (my_df$Temp - mean.Temp.All) / sd.Temp.All

head(my_df)

my_df

my_df[102, ]
```